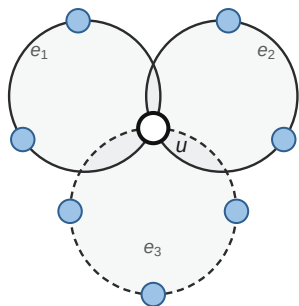


(a) Multiplex hypergraph

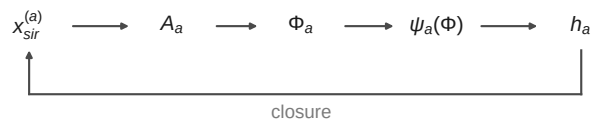
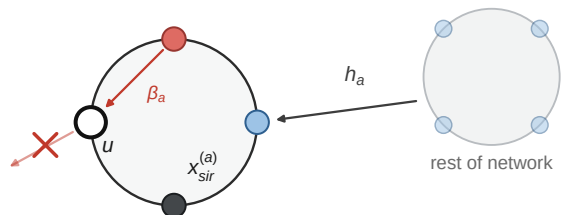


$$k^{(a)}(u) = 2$$

$$k^{(b)}(u) = 1$$

$$P(\mathbf{k}) \rightarrow \Psi, \psi_a$$

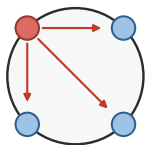
(b) Edge-based closure



$$\Rightarrow S(t), I(t), R(t), R(\infty)$$

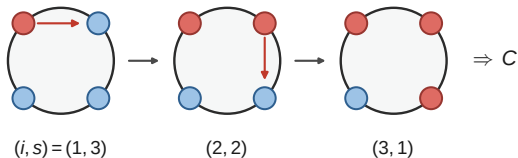
(c) Within-hyperedge factor C

direct



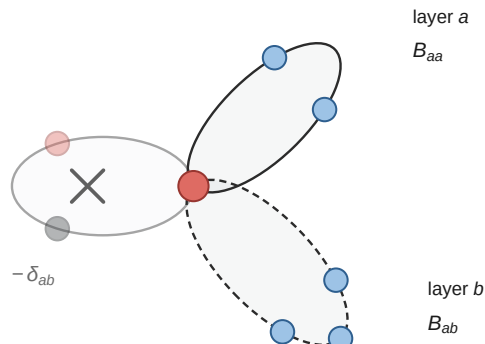
$$\Rightarrow (m-1)T$$

secondary



$$C \geq (m-1)T$$

(d) Excess hyperdegree B



$$K_{ab} = B_{ab} C_b$$

● susceptible
 ● infected
 ● recovered
 test node u

— layer a
 --- layer b